



— TECHNICAL DATA SHEET

# NN-Emul SMO

## Sorbitan Mono-oleate (SMO) – Industrial / Technical Grade



Industrial-grade non-ionic emulsifier – packed in 200 kg HDPE drums and 20 KL bulk tankers

| Document No.     | Revision | Issue Date | Product Code |
|------------------|----------|------------|--------------|
| NNPL/TDS/PROD-02 | 1.0      | 27-06-2026 | SMO-80       |

### 01 PRODUCT DESCRIPTION

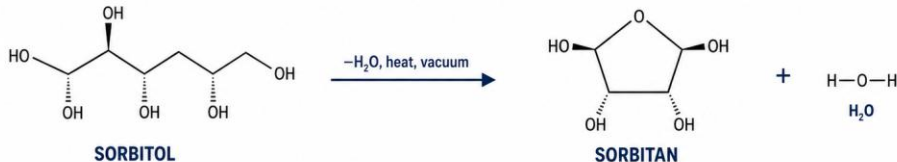
NN-Emul SMO is an industrial / technical grade Sorbitan Mono-oleate (SMO), a non-ionic surfactant produced by the esterification of sorbitan with oleic acid. It is an amber to brown viscous oily liquid with a mild characteristic odour and a low hydrophilic-lipophilic balance ( $HLB \approx 4.3$ ), which makes it a highly effective water-in-oil (W/O) emulsifier.

The product is oil-soluble and water-dispersible. It reduces interfacial tension between oil and water phases, forming a stable interfacial film that prevents coalescence and phase separation. It is commonly used on its own as a W/O emulsifier or in combination with high-HLB co-emulsifiers (such as Polysorbate 80) to build oil-in-water (O/W) systems at a targeted HLB.

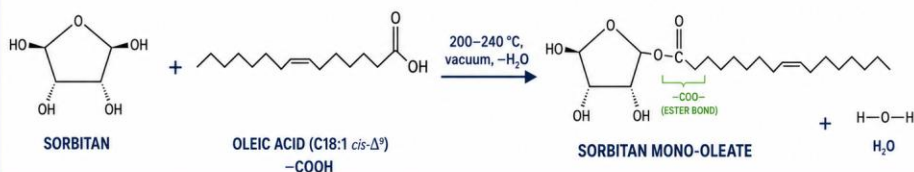


## TWO-STEP SYNTHESIS OF SORBITAN MONO-OLEATE

### 1 DEHYDRATION



### 2 ESTERIFICATION



*Two-step synthesis: dehydration of sorbitol to sorbitan, then esterification with oleic acid*

**Manufacturing:** Sorbitol is dehydrated to sorbitan, which is then esterified with oleic acid under controlled temperature (200–240°C) and vacuum; the water of reaction is removed by vacuum distillation, followed by neutralisation, filtration, cooling, quality testing and packaging.



*Reactor-based esterification plant for sorbitan ester manufacture*

## 02 CHEMICAL IDENTITY



| Attribute            | Value                                               |
|----------------------|-----------------------------------------------------|
| Chemical name        | Sorbitan mono-oleate (sorbitan ester of oleic acid) |
| Common / trade names | SMO, Sorbitan oleate, Sorbitan monooleate           |
| CAS number           | 1338-43-8                                           |
| EINECS / EC number   | 618-490-5                                           |
| INS / E number       | E494                                                |
| Molecular formula    | C <sub>24</sub> H <sub>44</sub> O <sub>6</sub>      |
| Molecular weight     | ~428.6 g/mol                                        |
| Surfactant class     | Non-ionic                                           |

### 03 KEY FEATURES & BENEFITS

- **Strong W/O emulsifier** — Low HLB (≈4.3) makes it highly effective at stabilising water-in-oil emulsions.
- **Versatile co-emulsifier** — Pairs with high-HLB surfactants (e.g. Polysorbate 80) to reach a precise target HLB for O/W systems.
- **Oil-soluble** — Soluble in vegetable and mineral oils and common organic solvents (ethanol, IPA, ethyl acetate).
- **Wetting & dispersing** — Acts as a wetting agent, dispersant, lubricant and viscosity/texture modifier.
- **Stable & compatible** — Compatible with a wide range of acids, bases and salts under normal conditions.
- **Biodegradable** — Readily biodegradable; favourable environmental profile for industrial use.

### 04 SPECIFICATION & TYPICAL PROPERTIES

| Property              | Specification                                                          | Method / Unit |
|-----------------------|------------------------------------------------------------------------|---------------|
| Appearance            | Amber to brown clear viscous oily liquid                               | Visual        |
| Odour                 | Mild, characteristic                                                   | —             |
| Acid value            | ≤ 8 mg KOH/g                                                           | Titration     |
| Saponification value  | 145 – 160 mg KOH/g                                                     | Titration     |
| Hydroxyl value        | 193 – 210 mg KOH/g                                                     | Titration     |
| Water content (KF)    | < 1.0 %                                                                | Karl Fischer  |
| HLB value             | ≈ 4.3                                                                  | Calculated    |
| Density @ 25°C        | 0.99 – 1.05 g/cm <sup>3</sup>                                          | ASTM D1475    |
| Pour / solidification | Pours at ambient; sets near ~10–12°C                                   | —             |
| Flash point           | > 110°C                                                                | Closed cup    |
| Solubility            | Oil-soluble; water-dispersible; soluble in ethanol, IPA, ethyl acetate | —             |
| Ionic nature          | Non-ionic                                                              | —             |
| Residue on ignition   | ≤ 0.5 %                                                                | Gravimetric   |

### 05 TYPICAL APPLICATIONS



- **Emulsion explosives** — W/O emulsifier in emulsion-explosive matrices — a well-established industrial application of sorbitan mono-oleate.
- **Agrochemicals** — Emulsifier / dispersant in emulsifiable concentrates (EC), suspension concentrates and oil-based crop-protection formulations.
- **Drilling fluids & oilfield** — Emulsifier and additive in oil-based drilling fluids and oilfield systems.
- **Lubricants & metal-working** — Emulsifier and lubricity additive in industrial lubricants and metal-working fluids.
- **Textiles & leather** — Softening, lubricating and antistatic auxiliary; fibre and leather processing.
- **Paints, inks & coatings** — Wetting and dispersing aid for pigments and fillers.

**Dosage:** Use level is formulation-specific and determined by the required HLB and emulsion type. Add to the oil phase and blend with mild heat (typically 40–60°C) until uniform.

## 06 PACKAGING, STORAGE & SHELF LIFE

| Item       | Detail                                                                                                                          |
|------------|---------------------------------------------------------------------------------------------------------------------------------|
| Packaging  | 200 kg HDPE drums; 20 KL bulk tanker                                                                                            |
| Storage    | Store in a cool, dry, well-ventilated area away from heat, ignition sources and direct sunlight; keep containers tightly closed |
| Shelf life | 24 months from date of manufacture in unopened original packaging                                                               |
| Handling   | If material has thickened in cold conditions, warm gently and mix before use; avoid prolonged exposure to air                   |

## 07 HEALTH, SAFETY & ENVIRONMENT

This product is not classified as hazardous under GHS for normal industrial handling. It is combustible but not readily ignitable, with a high flash point. Wear gloves and eye protection; avoid contact with eyes and prolonged skin contact. Spilled material is slippery. Refer to the accompanying Safety Data Sheet (SDS) for full first-aid, handling, spill and disposal guidance.

**Disclaimer:** The information in this TDS is given in good faith and is based on data believed to be reliable. NilayNarayan Polychem LLP makes no warranty, express or implied, as to its accuracy or completeness. Users must satisfy themselves as to the suitability and safe handling of the product for their own conditions and must comply with all applicable laws and regulations.